

Class 03 — Homework

Bong Study Hub — Foundation Batch 2026 · Class 03: Understanding Data

From the Real World → Information → Data

Estimated time: 30–45 minutes **Tools needed:** pen and paper, or a simple document — nothing else **Prerequisite:** Class 03. No coding, no software, no statistics, and no prior AI/ML knowledge required.

What This Assignment Is About

In class, we opened a box Class 02 left closed: **data**. We saw that data is just recorded information about something — not only numbers — and that it comes from the real world through observation and recording:

REAL WORLD → OBSERVATION → DATA

We looked at examples, datasets, rows, columns, features, and labels, and we saw that more data isn't automatically better data, and that useful data always depends on the question being asked.

This assignment asks you to **use these ideas, not recite them**. There is no coding, no math beyond everyday counting, and no single "correct sentence" to memorize — if your reasoning makes sense, your answer is good, even if it doesn't use the exact words from class.

Instructions

- Answer in **your own words**. Short answers are enough unless a question asks you to explain your reasoning.
- Several questions are **open-ended** — there may be more than one reasonable answer. You will not be marked down for a sensible answer that isn't the one your instructor happened to think of.
- **No coding is required**, and none should appear in your answers.
- You may refer to your **Student Notes** while answering.
- Where a question asks you to invent an example, feel free to make one up — you do not need to collect any real information from classmates.
- If you're unsure, **explain your reasoning anyway** rather than leaving a blank — partial thinking is worth more than an empty space.

Part A — Core Concepts

Question 1 — What Is Data? (2 marks)

In your own words, answer: **"What is data?"**

You do not need to reproduce an exact textbook definition — explain it the way you'd explain it to a friend who missed class.

Question 2 — Data Is Not Just Numbers (2 marks)

Give **three examples of data that are NOT numbers**. For each one, briefly state what real-world thing it represents.

(You don't need to use the exact examples from class.)

Question 3 — From Reality to Data (3 marks)

Scenario: You look outside and notice that it is raining.

Fill in the chain below for this scenario:

REAL WORLD: _____

OBSERVATION: _____

RECORDED DATA: _____

Then answer: **What makes the observation become data?**

Question 4 — Representation of Reality (2 marks)

Suppose a canteen is crowded, and someone records "80% full."

Is "80% full" the canteen itself? Explain why or why not.

Part B — Examples and Datasets

Question 5 — Read the Table (4 marks)

Student	Year	Attendance	Marks
A	1	82%	76
B	1	91%	84
C	1	68%	61

- a. What does one row represent?
- b. What does the **Attendance** column represent?
- c. How many examples are in this dataset?
- d. Why can the whole table be called a **dataset**?

Question 6 — Build Your Own Tiny Dataset (3 marks)

Imagine you are collecting information about college students. Invent a **tiny dataset** with:

- **3 examples** (rows)
- **3–4 columns**

Pick any theme you like — study habits, commute, daily routine, hobbies, or anything else. **You may invent fictional examples** — no need to collect real information from classmates.

Then answer:

What does one row in your table represent? What does one column represent?

Part C — Features and Labels

Question 7 — Feature or Label? (3 marks)

Email	Sender	Time	Label
"Win free money!"	Unknown	2:14 AM	Spam
"Meeting at 3 PM"	Colleague	9:02 AM	Not Spam

- Name two pieces of information that **describe** the email.
- What is the **label** for the first email?
- What is the **label** for the second email?
- Does **every** dataset necessarily have a label?

Part D — Thinking About Useful Data

Question 8 — The Canteen Problem (3 marks)

Suppose the college wants to answer: "**Can we predict when the canteen will be crowded?**"

List **five** pieces of information that might be useful.

Then pick **one** item from your list and explain why it might help.

Question 9 — Change the Question (2 marks)

Now the question changes to: **"Will a student like this movie?"**

Name **three** kinds of information that might be useful for this new question.

Then answer: **Why wouldn't the canteen data necessarily be useful for this question?**

Part E — Data Quality and Reality

Question 10 — More Data ≠ Better Data (2 marks)

Imagine two datasets:

- **Dataset A:** 1,000,000 examples, but many are incorrect or repetitive.
- **Dataset B:** 1,000 examples, accurate and covering useful variety.

Which dataset would you rather use, and why? Explain in 2–4 sentences.

Question 11 — Spot the Mess (2 marks)

Student	Age	City
A	18	Kolkata
B	"eighteen"	kolkata
C	missing	Calcutta

Identify at least two problems in this dataset.

Part F — Final Synthesis

Question 12 — Connect Class 02 and Class 03 (2 marks)

Fill in the blank:

REAL WORLD
↓

↓
DATA
↓
LEARNING
↓
MODEL
↓
PREDICTION

Then, in **one or two sentences**, answer: **What did Class 03 add before the DATA step we learned about in Class 02?**

Optional Challenge — Not Scored (up to 3 bonus marks)

This section is entirely optional and will not affect your base grade.

Suppose you want to predict **whether a college student will attend an event**.

- What information might be genuinely useful?
- What information might *look* interesting but actually be irrelevant?

Explain your reasoning. There is no single predetermined correct list — this is a chance to practice turning a question into a plan for what to observe.

Submission Format

Choose whichever is easiest for you:

- **Option A:** Handwrite your answers, then photograph or scan them.
- **Option B:** Type your answers in a simple document.
- **Option C:** Submit as a simple PDF.

Please label each answer with its question number.

Before You Submit — Self-Check

- ☐ I answered all 12 questions in my own words.
- ☐ Where unsure, I explained my reasoning instead of leaving a blank.
- ☐ I did not use code or programming syntax anywhere.
- ☐ I invented fictional examples rather than collecting real information about classmates.
- ☐ I reviewed my work before submitting.